American Computer Science League

2023 Finals ● Short Problems ● Intermediate/Classroom Division

1. Boolean Algebra

Simplify the following Boolean expression:

$$(\overline{B + C})(\overline{A}B + \overline{C})(\overline{A + B + C})$$

A. $A\,\overline{B}\,\overline{C}$
B. $\overline{A}\,\overline{B}\,\overline{C}$
C. $\overline{A}\,B\,\overline{C}$
D. $\overline{A}\,\overline{B}\,C$
E. None of the above

2. Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{A(\overline{BC} + C)}\ (\overline{B\overline{C}} + A)$$

A. 1
B. 3
C. 4
D. 6
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

NOT (010111 XOR 011011 OR 001010 AND 111010)

A. 001100
B. 110001
C. 001010
D. 001110
E. None of the above

4. Bit-String Flicking

How many different values of X (a bit-string of 5 bits) make the following equation TRUE?

(LSHIFT-1 (RCIRC-2 11011)) OR X = 11110

A. 0
B. 2
C. 4
D. 8
E. None of the above

5. Recursive Functions

Find $f(11)$ if given:

$$f(x) = \begin{cases} 3 & \text{if } x = 1 \\ 2 \cdot f(x-1) - 1 & \text{if } x > 1 \end{cases}$$

A. 2047
B. 2048
C. 2049
D. 2050
E. None of the above

6. Recursive Functions

Find $f(24, 6)$ if given:

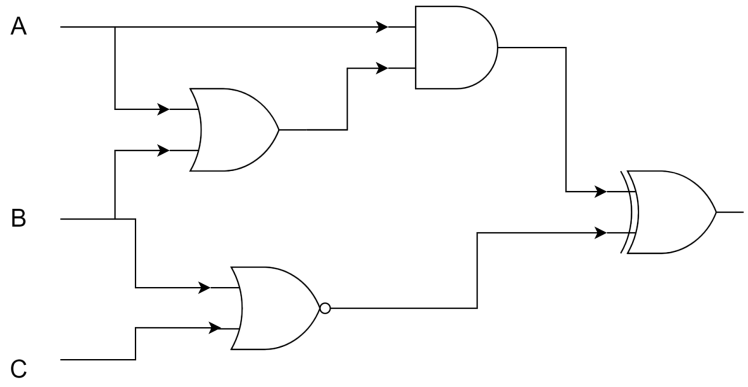
$$f(x,y) = \begin{cases} f\left(\left[\frac{x}{2}\right], y-1\right)+3 & \text{if } x \geq 10 \\ 2 \cdot f\left(x+1, \left[\frac{y}{3}\right]\right) & \text{if } 5 < x < 10 \\ x \cdot y & \text{if } x \leq 5 \end{cases}$$

Note: [a] is the greatest integer less than or equal to a

A. -32
B. -29
C. -26
D. -5
E. None of the above

7. Digital Electronics

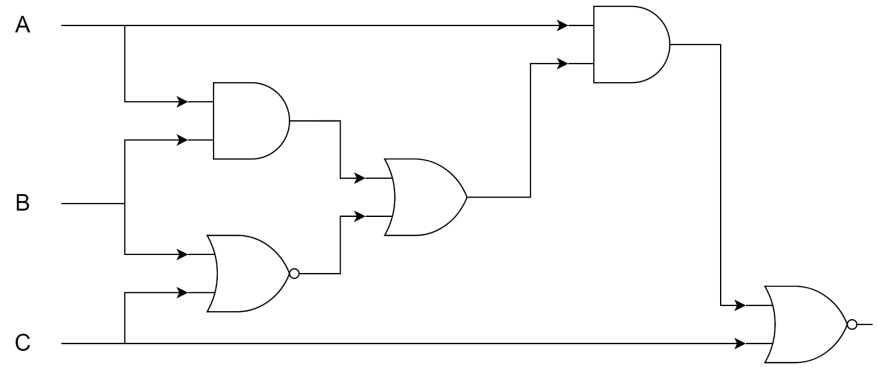
How many ordered triples make the following circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. None of the above

8. Digital Electronics

Simplify the following Boolean expression represented by the digital circuit:



A. $A\,C$
B. $\overline{A}\,C$
C. $A\,\overline{C}$
D. $\overline{A}\,\overline{C}$
E. None of the above

9. Prefix-Infix-Postfix

Define a \$ b = $(a-b)^2$. Evaluate the following prefix equation, if all numbers are single digits:

+ \$ 7 3 − * \$ 9 7 3 5

A. 33
B. 23
C. 16
D. 7
E. None of the above

10. Prefix-Infix-Postfix

Convert this infix expression to postfix:

$$\frac{(A+B)(A-B)}{B} \quad _ \quad \frac{B(4-A)}{A}$$

A. AB+AB*−B/B4A−*A/−
B. AB+AB−*B/B4A*−A/−
C. AB+AB−*B/B4A−*/A−
D. AB+AB−*B/B4A−*A/−
E. None of the above

11. Computer Number Systems

Convert 52723_8 to hexadecimal.

A. ABD3
B. 55D3
C. 56D3
D. A5D3
E. None of the above

12. Computer Number Systems

What is the value (in base 10) of the hex number 1ABC?

A. 6604
B. 6588
C. 6832
D. 6844
E. None of the above

13. Data Structures

Build a binary search tree for:

RHINOCEROS

What is the internal path length of the tree?

A. 15
B. 26
C. 29
D. 33
E. None of the above

14. Data Structures

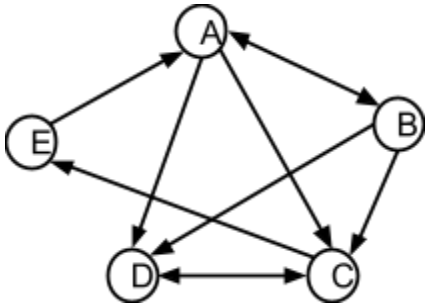
Given some method used for distributing basketballs to people arriving at the gym for recreational play, which of the following could be simulated by a stack?

a) a check-out counter where each person arriving gets whatever ball was least recently used
b) a bin of balls where each person grabs one closest to the top
c) a trough diagonally slanted toward the floor so players get the one closest to the floor and return it at the top of the trough
d) a spring-loaded device where the ball on top is retrieved and when returned each ball is put on top by pushing the other balls further down

A. a
B. b
C. c
D. d
E. None of the above

15. Graph Theory

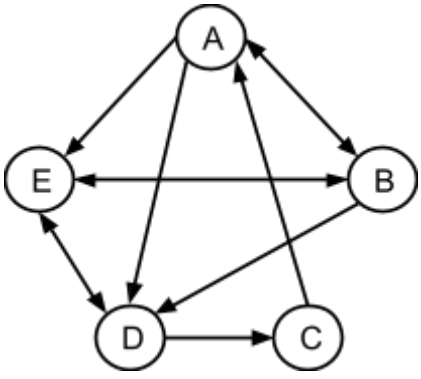
Given the directed graph below, how many pairs of vertices do not have paths of length 2?



A. 8
B. 9
C. 10
D. 11
E. None of the above

16. Graph Theory

How many cycles from A are there in this directed graph?



A. 6
B. 7
C. 8
D. 9
E. None of the above

17. What Does This Program Do?

Given that array **arr** contains the 10 values below, what is the final value of **count**?

5	8	6	2	9	3	7	0	1	4

```
count = 0
for x = 0 to 8
     for y = x+1 to 9
          if x == arr(y) then
               z1 = x
               z2 = y
               while z1 < z2
                    t = arr(z1)
                    arr(z1) = arr(z2)
                    arr(z2) = t
                    count = count + 1
                    z1 = z1 + 1
                    z2 = z2 - 1
               end while
          end if
     next y
next x
```

A. 16
B. 17
C. 18
D. 19
E. None of the above

18. LISP

After the following LISP program is run, what is the value of the last expression?

(SETQ M '(1 (2 3) (4 (5) (6 7))))
(CDR (REVERSE (CAR (REVERSE (CDR M)))))

A. ((7 6) (5) 4)
B. ((5) 4)
C. (5 4)
D. ((6 7) (5))
E. None of the above

19. FSAs and Regular Expressions

Given the following Regex commands:

\d Matches any digit.
\D Matches any character that is not a digit.

Welcome to the ACSL Maternity Ward! If parents name their child so that it satisfies the following regular expression, the hospital bill will be reduced by 50%. Which names get a discount?

[^bru] [a-iro] [\D]* [a-n]* [^mr] [\d]*

a) asher	b) aurora	c) c++	d) jo-ann
e) trish	f) noah2	g) a1b2c3	h) violet
i) william	j) eleanor	k) yanni	l) lee

A. d, e, f, h, k, l
B. d, e, i, j, k
C. c, d, j, k, l
D. b, d, f, g, k, l
E. None of the above

20. Assembly Language

What is printed when the following program is run?

```
NUM       DC        20
M         DC        0
TOP1      LOAD      NUM
          SUB       =2
          BL        STOP
          LOAD      =2
          STORE     FACT
TOP2      LOAD      NUM
          DIV       FACT
          STORE     QUOT
          MULT      FACT
          SUB       NUM
          BE        NEXT
          LOAD      FACT
          ADD       =1
          STORE     FACT
          BU        TOP2
NEXT      LOAD      FACT
          SUB       M
          BL        CONT
          LOAD      FACT
          STORE     M
CONT      LOAD      QUOT
          STORE     NUM
          BU        TOP1
STOP      PRINT     M
          END
```

A. 2
B. 3
C. 5
D. 7
E. None of the above